Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/12/2023
PI Name:	Riaz, Amir Ph.D.		
Project Title:	Pool Boiling Heat Transfer Mechanisms in Low Gravity: Numerical Experiments of MABE and NPBX Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017/2019 Physical Sciences NNH17ZTT001N-17PSI-F: Use of the NASA Physical Sciences Informatics System – Appendix F
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2024 per PI (Ed., 11/27/23) NOTE: End date changed to 11/30/2023 per NSSC information (Ed., 11/22/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balaras, Ilias Ph.D. (George Washington University) Kim, Jungho Ph.D. (University of Maryland, College Park)		
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Task Description:	The key challenge in developing a proper understanding of nucleate boiling in thermal systems for space environments is the accurate characterization of bubble dynamics and associated heat transfer mechanisms over a wide range of wall superheat and liquid subcooling conditions. To address this challenge, we are proposing well-characterized numerical experiments using high fidelity Direct Numerical Simulation for investigating the dynamics of nucleate boiling across a range of gravity conditions. Our numerical experiments will recreate the conditions of the Microheater Array Boiling Experiment (MABE) and Nucleate Pool Boiling eXperiment (NPBX) that were flown on the International Space Station (ISS) in 2011 within the Boiling Experiment Facility (BXF). MABE provided local heat transfer and visualization of pool boiling heat transfer behavior in a microgravity environment. The focus of NPBX was the investigation of how bubbles grow and merge in microgravity. The main finding was that heat fluxes for steady state nucleate boiling were found to be substantially lower than what was expected on the basis of low gravity data obtained on Earth. The numerical solver that we will use to simulate MABE conditions has been recently validated by comparing the heat flux from our fully 3-D simulations to MABE data for a representative sample of wall superheat and liquid subcooling conditions. Details about our numerical methods and validation case studies have been recently published (Akash Dhruv, Elias Balaras, Amir Riaz, and Jungho Kim. A formulation for high-fidelity simulations of pool boiling in low gravity. International Journal of Multiphase Flow, 120:103099, 2019). These case studies demonstrate that our solver is able to resolve fully the complex interactions of liquid and vapor phases involving phase transition, interphase mass flux, and contact line dynamics. Using the data from the simulations, we have been able to connect the dynamics of bubble nucleation and growth to the corresponding heat flux measurements in MABE data. Such detailed characterizations of the flow field have already started to shape and improve our understanding of the underlying bubble dynamics. In the proposed research, we aim to systematically probe the MABE database now at a deeper level to extract fundamental information on: (i) how is heat flux partitioned into contributions arising from nucleate boiling and natural convection? (ii) what is the role of contact angle in affecting the force of wall adhesion? and (iii) how much heat transfer occurs through microlayers in low gravity settings? These questions will be addressed to qualitatively different regimes of bubble dynamics under various conditions of gravity, wall superheat, and liquid subcooling. The answers to these questions, investigated on the basis of a validated solver using MABE data, would extend the general understanding of pool boiling in low gravity settings. The use of the MABE database is instrumental to this endeavor. It will be used to provide the reference dataset that connects heat flux, and in some cases even visualization of nucleate boiling, to different levels of gravity, wall super heat, and subcooling. This will allow a unique opportunity for a quantitative assessment of potential improvements in the design of thermal systems in space applications. The proposed research is expected to complement NASA's ongoing efforts, in addition to advancing the science of heat flux partitioning, contact line dynamics, and the role of microlayers for nucleate boiling in low gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Electronic devices require effective and efficient means of thermal management to achieve performance targets. The use of latent heat removal via the process of phase change, such as nucleate and pool boiling, is an extremely effective method of thermal management. This method relies primarily on the dominance of buoyancy over capillary forces to maintain high heat flux at the heated surface. The proposed research will study the influence of major parameters on the behavior of boiling heat transfer. The resulting data and analysis will help improve the thermal management of electronic devices, enabling the achievement of higher performance goals.
Task Progress:	Numerical simulations were carried out using experimental conditions of the Microheater Array Boiling Experiment (MABE) dataset. Simulations were performed on the NASA Advanced Supercomputing (NAS) system. Each simulation used around one thousand cores and ran for approximately 200 hours. The research activities fall into the following two main categories. Task 1. Study of the effect of dimensionless parameters on subcooled nucleate pool boiling Two sets of numerical experiments were carried out using FC-72 as a working fluid to match the working fluid of MABE dataset. In the first set, the effect on subcooling was studied by varying the bulk liquid temperature while maintaining all other parameters constant. For the second set, the Stefan number was held constant and the effect of changing the wall superheat and bulk liquid temperature was studied. All of the cases in the two groups demonstrate that liquid subcooling impacts the boiling curve, the heat flux, and bubble dynamics. An increase in the level of subcooling was observed to reduce wall heat flux due to the decrease in bubble departure frequency and reduction in bubble growth up to a threshold beyond which the trend was reversed. Bubble statistics, such as departure diameters obtained from the numerical solution, were compared with existing empirical correlations. The comparison shows that numerical data fall within the range of values derived from the correlations. However, empirical correlations were generally found to be imprecise because they neglect the effect of bubble departure rate on subcooling. Task 2. Study of the effect of heterogenous wettability surfaces on the thermal performance of nucleate boiling. Numerical experiments of pool boiling were performed to help identify the key physical mechanisms that are responsible for enhancing heat transfer on biphilic surfaces. A biphilic parameter, defined as the ratio of hydrophobic surface area to total surface area (A*=" Hydrophobic area / Total area"), was varied and its effect on the rate of heat transfer was investigated. Three cases of A^*= 0.1, 0.25 and 0.75 were analyzed. The highest cumulative heat transfer rate was observed for A^*=0.25. The thermal-fluid dynamic features of pool boiling observed in the simulation were examined in detail to understand and interpret these observations. In particular, a parachute-like (PCH) vapor bubble structure is formed, which is primarily responsible for the enhancement of heat flux on biphilic surfaces for some intermediate value of A^*. The formation, development, and evolution of this structure is explained in detail. It is noted that these structures directly cause the formation of high heat flux regions, and their contribution to the total heat rate is the maximum for an intermediate case of A^*=0.25, for the set of parameters considered in this case.
Bibliography Type:	Description: (Last Updated: 04/23/2026)

Abstracts for Journals and Proceedings	Youssoufi S, Riaz A, Balaras E. "Pool boiling in Earth and micro-gravity during sub-cooling temperatures." 75th Annual Meeting of the APS Division of Fluid Dynamics, Indianapolis, IN, November 20-22, 2022. Bulletin of the American Physical Society. 2022 Nov;67(19):Abstract: A34.00001. https://meetings.aps.org/Meeting/DFD22/Session/A34.1 , Nov-2022
Papers from Meeting Proceedings	Youssoufi S, Lentnera A, Riaz A, Balaras E. "A study of subcooled pool boiling using direct numerical simulations." 4th International Conference on Fluid Flow and Thermal Science (ICFFTS 2023), Lisbon, Portugal, December 7-9, 2023. Proceedings of the 4th International Conference on Fluid Flow and Thermal Science (ICFFTS 2023), Lisbon, Portugal, December 7-9, 2023. Paper 169. http://dx.doi.org/10.11159/icffts23.169 , Dec-2023